

An AI-Enabled Early Warning System for Sepsis-3 Onset Prediction Using Temporal Deep Learning on the MIMIC-IV Clinical Database

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Abstract—Sepsis remains one of the most pressing challenges in modern critical care. Despite decades of clinical research, the tools doctors use at the bedside to identify patients at risk, most notably the quick Sequential Organ Failure Assessment (qSOFA) score, are surprisingly unreliable, missing more than 62% of at-risk patients before they deteriorate. In this paper, we ask a simple but important question: can we do meaningfully better using AI?

We present a complete, reproducible machine learning pipeline for predicting Sepsis-3 onset six hours before clinical recognition, trained on the full MIMIC-IV clinical database (94,458 ICU stays across 364,627 patients; 4,571,933 hourly feature rows; 37,994 positive labels, 0.83% prevalence). We systematically compare four model families, Logistic Regression, XGBoost, Random Forest, and a Long Short-Term Memory (LSTM) recurrent neural network, using patient-grouped cross-validation to prevent data leakage throughout.

Our key finding demonstrates that a two-layer LSTM trained on 12-hour sliding windows of structured vital signs and laboratory data achieves an AUROC of 0.9761 (AUPRC: 0.6355; Sensitivity: 78.98%; Precision: 57.18%), a result competitive with previously reported multi-center clinical ML systems and a substantial improvement over the best 5-fold-GroupKFold-validated static model on the same data (XGBoost, AUROC: 0.809). A clinically important observation is the reduction in false-alarm burden: XGBoost's optimal operating point generates approximately 14.4 false alarms per true positive, while the LSTM reduces this to approximately 0.75 false alarms per true positive, roughly a 19-fold improvement that could meaningfully reduce alert fatigue in real ICU settings.

Explainability analysis using SHAP identifies respiratory organ dysfunction and elevated lactate as the two dominant predictors, consistent with established sepsis pathophysiology. We also evaluate a multimodal extension using PCA-derived secondary features, which yields a negligible improvement over the tabular model alone (AUROC +0.0002), a null result we report transparently and interpret as motivation for future work using genuine clinical-note embeddings rather than a proxy signal. Finally, we propose a four-layer hierarchical surveillance architecture that extends individual patient prediction to hospital-level anomaly detection, regional outbreak monitoring, and national infectious disease intelligence. All results reported here are retrospective and single-center; external validation and a fully unified cross-

validation protocol with confidence intervals remain important next steps before any clinical deployment is considered.

Code and trained models are publicly available at [\[https://github.com/healthai-2026/Infectious-Disease-Triage\]](https://github.com/healthai-2026/Infectious-Disease-Triage).

Index Terms—Sepsis-3 prediction, MIMIC-IV, LSTM, recurrent neural network, early warning system, SHAP explainability, clinical decision support, temporal deep learning, intensive care unit, electronic health records, infectious disease surveillance.

I. INTRODUCTION

SEPSIS remains one of the leading causes of mortality among critically ill patients worldwide. Despite continuous advances in intensive care medicine, early recognition of sepsis continues to be a major clinical challenge. The condition develops when the body's response to an infection becomes dysregulated, leading to progressive organ dysfunction and, if left untreated, septic shock. Previous studies have shown that every hour of delay in administering effective antimicrobial therapy significantly increases the risk of mortality. Consequently, identifying patients at risk before obvious clinical deterioration occurs has become an important objective in critical care research.

Although several bedside assessment tools are routinely used in hospitals, their predictive capability remains limited. The quick Sequential Organ Failure Assessment (qSOFA) score, for example, evaluates only three physiological variables and reflects the patient's condition at a single point in time. While its simplicity makes it practical in emergency settings, it cannot capture the gradual physiological changes that often precede sepsis onset. In clinical practice, physicians frequently rely not only on current measurements but also on how these measurements evolve over time.

The widespread adoption of electronic health record (EHR) systems has created opportunities to develop data driven approaches for early disease prediction. Modern intensive care units continuously collect vital signs, laboratory investigations, medication records, and clinical observations throughout a patient's stay. Machine learning models can leverage these large scale longitudinal datasets to identify complex relationships that are difficult to recognise through conventional statistical methods.

Among recent machine learning approaches, recurrent neural networks, particularly Long Short Term Memory (LSTM) networks, have shown considerable promise for modelling sequential clinical data. Unlike conventional machine learning algorithms that depend on handcrafted summary statistics,

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LSTM networks retain temporal information and learn how physiological variables change over time. This capability makes them particularly suitable for early warning applications where disease progression is reflected through evolving clinical trends rather than isolated measurements.

In this study, we present an AI enabled early warning system for predicting Sepsis-3 onset six hours before clinical recognition using the MIMIC-IV database. Four machine learning approaches: Logistic Regression, Random Forest, XGBoost, and Long Short Term Memory (LSTM), are evaluated under a patient level validation framework to ensure a fair comparison. Besides predictive performance, the study also investigates model interpretability through SHAP analysis and discusses the practical challenges associated with deploying AI-based decision support systems in intensive care environments.

Beyond individual patient prediction, this work also outlines a hierarchical surveillance framework that extends hospital-level predictions to regional and national infectious disease monitoring. Although this broader framework remains conceptual, it highlights the potential of temporal machine learning models to support both clinical decision-making and public health surveillance.

A. Contributions

The primary contributions of this work are summarised as follows:

- 1) A complete preprocessing pipeline is developed to transform raw MIMIC-IV electronic health records into hourly Sepsis-3 prediction samples while adhering to the official clinical definition.
- 2) Four machine learning models: Logistic Regression, Random Forest, XGBoost, and LSTM are systematically evaluated using patient-level validation to prevent data leakage.
- 3) The proposed temporal deep learning model demonstrates the advantages of preserving sequential clinical information for early sepsis prediction.
- 4) SHAP based explainability analysis is performed to identify clinically meaningful predictors and improve model transparency.
- 5) A hierarchical AI surveillance framework is proposed to extend patient level prediction toward hospital, regional, and national infectious disease monitoring.

II. RELATED WORK

Many studies have tried to improve the early detection of sepsis. Earlier work mainly focused on clinical scoring systems. More recent studies have used machine learning and deep learning to analyse electronic health records. This section briefly reviews these approaches and explains how our work is different.

A. Clinical Scoring Systems

For many years the SIRS criteria were used to identify patients with sepsis. The system looked at signs such as body temperature heart rate respiratory rate and white blood cell

count. Although it was simple to use it often identified patients who did not actually have sepsis. This reduced its reliability in clinical practice [6].

To overcome these problems the Sepsis-3 guidelines introduced the SOFA score. Instead of looking only at signs of infection the SOFA score measures organ dysfunction across different body systems. Studies have shown that SOFA provides better prediction of poor patient outcomes than both SIRS and qSOFA [1], [5].

The qSOFA score was later introduced as a faster bedside screening tool. It uses only three clinical measurements and is easy to calculate without laboratory tests. However many studies have reported that qSOFA is useful mainly after patients become seriously ill. Its ability to predict sepsis several hours before clinical recognition is limited [4].

B. Machine Learning for Sepsis Prediction

The availability of electronic health records has encouraged researchers to apply machine learning to sepsis prediction. Traditional models such as Logistic Regression Random Forest and XGBoost have been widely used because they are simple to train and provide good baseline performance. These models learn patterns from patient data and often perform better than traditional clinical scoring systems.

As larger clinical datasets became available researchers started using deep learning models. Recurrent Neural Networks especially Long Short-Term Memory (LSTM) networks became popular because they can learn from sequential patient data. Instead of analysing one measurement at a time they learn how a patient's condition changes over several hours. Harutyunyan et al. showed that LSTM models perform better than traditional machine learning methods when analysing clinical time series data [10].

The PhysioNet Challenge 2019 also encouraged researchers around the world to develop early sepsis prediction models. Most of the successful approaches used either gradient boosting methods or recurrent neural networks which highlighted the importance of learning from temporal patient information [11].

C. Recent Developments

Recent research has focused on improving prediction accuracy while making AI systems more useful in hospitals. Dobariya et al. developed a machine learning based early warning system that reduced the time required for antibiotic treatment in real clinical settings [13]. Their work showed that AI can improve patient care when used alongside clinicians.

Chang et al. proposed a more advanced framework that combines federated learning temporal transformers and knowledge graphs [14]. Their approach achieved strong predictive performance while allowing hospitals to train models without sharing patient data. This is an important step towards privacy preserving healthcare AI.

Although these methods produce excellent results they also require more complex architectures larger computational resources and data from multiple hospitals. In comparison our work focuses on building a simpler and reproducible

pipeline using the MIMIC-IV dataset. We compare traditional machine learning models with an LSTM model under the same prediction task to better understand the value of temporal learning. We also include SHAP analysis to explain the model predictions and discuss how the system can be extended for future hospital and public health surveillance. Our results, while obtained on a different dataset and prediction horizon, are broadly consistent with the finding from the PhysioNet/CinC 2019 Challenge [11] that temporal, sequence-aware models tend to outperform models built purely from static summary features.

III. METHODS

A. Dataset

The experiments were carried out using the MIMIC-IV version 3.1 database [8]. MIMIC-IV is a large public electronic health record dataset collected from patients admitted to Beth Israel Deaconess Medical Center in Boston between 2008 and 2019. The complete v3.1 release contains 364,627 patients and 94,458 ICU stays; after applying our cohort inclusion and exclusion criteria, the pipeline produced the 4,571,933-row hourly feature matrix described in Section IV.

The database includes a wide range of clinical information such as patient demographics, vital signs, laboratory investigations, medication records, microbiology reports, and nursing observations. These records provide a detailed timeline of each patient's condition during their stay in the ICU, making the dataset suitable for developing early prediction models.

Access to the dataset was obtained through PhysioNet after completing the required CITI training and accepting the data use agreement [9]. Since all personal identifiers have been removed, the dataset is safe to use for research while protecting patient privacy.

B. Cohort Definition

Only adult patients aged 18 years or older were included in this study. Patients with ICU stays shorter than six hours were excluded because they do not provide enough clinical information for predicting sepsis several hours in advance.

Patients who already satisfied the Sepsis-3 criteria at the time of ICU admission were also removed. Since the objective of this work is early prediction, including patients who were already septic at admission would not provide useful training examples.

C. Defining Sepsis-3 in the Data

The Sepsis-3 clinical definition was used throughout this study to generate the target labels. According to this definition, sepsis is identified when suspected infection is followed by a rise of at least two points in the SOFA score [1], [7]. To apply this definition, the time of suspected infection and the hourly SOFA score were calculated for every patient.

TABLE I
SOFA SCORE COMPONENTS AND MIMIC-IV DATA SOURCES

Component	Variable	MIMIC-IV Table	Key IDs	Item
Respiratory	PaO ₂ /FiO ₂	labevents / chartevents	50821, 223835	
Coagulation	Platelets ($\times 10^3/\mu\text{L}$)	labevents	51265	
Liver	Bilirubin (mg/dL)	labevents	50885	
Cardiovascular	MAP + vasopressors	chartevents / in-putevents	220052, 221906	
Renal	Creatinine / Urine output	labevents / out-putevents	50912, 226559	
Neurological	Glasgow Coma Scale	chartevents	220739, 223900	

1) *Suspected Infection*: The suspected infection time (t_{SI}) was determined using microbiology culture orders and antibiotic prescriptions. Following the Sepsis-3 guidelines, an infection was considered suspected when antibiotic administration occurred within the accepted time window around a culture order.

The suspected infection time (t_{SI}) is defined as the earlier of a microbiological culture order and an antibiotic prescription, subject to the constraint that antibiotic administration falls within

$$[t_{culture} - 24h, t_{culture} + 72h]$$

as described by Seymour et al. [7]. Antibiotics were identified by matching prescription records with a curated list of commonly used antibiotic names. Drug names that could produce false matches were excluded to avoid incorrect labelling.

2) *SOFA Score*: After identifying suspected infection, the SOFA score was calculated every hour using structured clinical data available in MIMIC-IV. The score measures the function of six different organ systems and provides an overall assessment of organ failure.

Whenever more than one measurement was available, the clinically preferred value was selected. For example, arterial oxygen measurements were given priority over non-invasive measurements whenever possible. Similar decisions were followed for the remaining variables to maintain consistency across all patients.

The admission SOFA score was used as the baseline, and patients were considered septic when their SOFA score increased by two or more points after the time of suspected infection. Physiologically impossible measurements were removed before calculating the hourly scores.

3) *Sepsis-3 Onset and Prediction Task*: A patient is labeled Sepsis-3 positive at time t_{onset} when their SOFA score has increased by ≥ 2 points from ICU admission baseline in the presence of suspected infection. We frame prediction as binary classification: for each hourly time point t ,

$$y_t = \begin{cases} 1 & \text{if } 0 < (t_{onset} - t) \leq 6 \text{ hours} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Time points at or after t_{onset} are excluded to prevent the model from being trained on post-onset observations, which would constitute label leakage.

D. Feature Engineering

For each prediction time point t , we extract features from a 6-hour lookback window $[t - 6h, t]$. For each of seven vital signs, we compute six summary statistics:

$$\phi_v = \{\mu, \sigma, x_{\min}, x_{\max}, x_{last}, \beta\} \quad (2)$$

where β is the linear regression slope over the window, a simple but effective measure of whether a vital sign is trending up or down. The full feature set comprises 62 core features: 7 SOFA components; 42 vital-sign statistics (7 vitals $\times$ 6 statistics); 13 most-recent lab values (glucose, potassium, sodium, creatinine, chloride, BUN, hematocrit, bicarbonate, platelets, hemoglobin, WBC, bilirubin, lactate); and 2 demographic features (age, gender).

SOFA sub-scores are included as input features. This creates partial overlap with the SOFA-based outcome label: because the label is itself defined by a SOFA increase, a model that can observe the current SOFA trajectory has partial access to information that mechanically anticipates the label, which can inflate apparent performance relative to a model restricted to raw vitals and labs alone. This is an approach consistent with prior work in this area, and we discuss it transparently in Section VII, where we also note training without SOFA features as a planned complementary experiment.

E. Models

Logistic Regression, XGBoost, and Random Forest were selected as baselines because they represent the three model families most commonly reported in the sepsis prediction literature [11], [12] and provide a spectrum of inductive bias, from fully linear to non-linear ensemble methods, against which the temporal LSTM can be meaningfully compared.

1) *Logistic Regression*: A standard ℓ_2 -regularized logistic regression ($C = 0.1$) with balanced class weighting and standardized inputs via a median-imputation, scaling, and classification pipeline.

2) *XGBoost*: A gradient-boosted tree ensemble with shallow trees ($\text{max_depth}=1$), strong regularization ($\alpha = 1.0$, $\lambda = 5.0$), and per-fold scale_pos_weight (≈ 119) to handle the 0.83% positive rate.

3) *Random Forest*: A bagged ensemble with 200 trees, maximum depth 8, and balanced class weighting, evaluated on a patient-safe 80/20 split.

4) *LSTM*: Our primary model is a two-layer LSTM that processes 12-hour sliding windows of hourly data (stride = 3 h):

$$\mathbf{h}_t = \text{LSTM}_\theta(\mathbf{x}_t, \mathbf{h}_{t-1}) \quad (3)$$

$$\hat{y} = \sigma(\mathbf{W} \cdot \text{LayerNorm}(\mathbf{h}_{12}) + b) \quad (4)$$

with input size 7 (one value per vital sign at each hourly step), hidden size $d = 128$, dropout $p = 0.3$, a final linear layer followed by a sigmoid activation producing a

scalar sepsis probability, gradient clipping at norm 1.0, and Adam optimization ($\eta = 10^{-3}$, weight decay 10^{-4}), trained with BCEWithLogitsLoss under positive class weighting for a maximum of 5 epochs per fold with batch size 128. From 4.57M hourly rows, we construct 354,840 sequences of shape (12×7) , achieving a 4.85% positive rate at the sequence level, substantially higher than the 0.83% row-level rate because each positive stay contributes multiple high-risk windows near sepsis onset.

F. Multimodal Ablation

To assess whether a secondary feature modality could improve performance without requiring clinical-note access, we implemented a proxy pipeline using PCA-derived embeddings of the existing numeric features as a stand-in secondary signal, and trained three XGBoost variants: tabular-only, PCA-only, and tabular+PCA combined. This experiment is intended primarily to validate the ablation infrastructure ahead of future work with genuine clinical-note embeddings.

G. Evaluation Protocol

All models use Group Cross-Validation grouped by `subject_id` where noted (5-fold for Logistic Regression and the primary XGBoost model; 2-fold for LSTM), or a patient-safe 80/20 split (Random Forest, ablation XGBoost variants), ensuring no patient's rows appear in both training and evaluation data under either protocol. Decision thresholds are tuned on out-of-fold (or held-out) predictions to maximize F1-score. We report AUROC, AUPRC, sensitivity, specificity, precision, and F1, emphasizing AUPRC and precision because, at a 0.83% positive rate, AUROC alone can be misleadingly optimistic.

H. Explainability

SHAP values were computed for the final XGBoost model using `TreeExplainer` on a stratified sample of 5,000 rows (42 positive), providing consistent feature-attribution scores grounded in cooperative game theory. Feature importance is reported globally (mean absolute SHAP value across the sample) in Table III and Fig. 7, and at the level of individual predictions via the dependence and waterfall plots in Fig. 8.

IV. RESULTS

This section presents the performance of all four machine learning models on the MIMIC-IV dataset. The results are compared using several evaluation metrics to understand how well each model predicts Sepsis-3 before clinical onset.

A. Dataset Statistics

After preprocessing, the dataset contained 4,571,933 hourly patient records generated from ICU stays in the MIMIC-IV database. Among these records, 37,994 were labelled as positive for Sepsis-3, while the remaining records represented non-sepsis cases. This resulted in a highly imbalanced dataset, with positive samples making up only about 0.83% of the total data.

Such class imbalance is common in medical prediction problems and makes evaluation more challenging. For this reason, multiple evaluation metrics were used instead of relying only on prediction accuracy.

B. Model Performance

Table II compares the performance of all four machine learning models along with the clinical qSOFA baseline.

The traditional machine learning models showed reasonable performance, but clear differences were observed among them. Logistic Regression produced stable results and served as a useful baseline because of its simplicity. Random Forest and XGBoost achieved better discrimination than Logistic Regression by learning more complex relationships from the clinical features.

The LSTM model achieved the best performance across almost all evaluation metrics. Unlike the other models, it learns directly from sequential patient observations rather than relying only on summary statistics. This allows the model to capture changes in a patient's condition over time, which is especially important for predicting sepsis before obvious clinical deterioration occurs.

Overall, the results suggest that preserving temporal information provides a clear advantage for early sepsis prediction.

C. Multimodal Ablation Results

To evaluate whether a secondary feature modality contributes additional predictive information, three XGBoost variants were trained: tabular features only, PCA-derived features only, and a combination of both. Fig. 6(a) shows the performance of the PCA-only variant, and Fig. 6(b) shows the performance of the combined tabular+PCA variant. Because the PCA-derived signal was constructed from the same underlying numeric features rather than an independent data source, the combined model produced only a marginal change relative to the tabular-only model, while the PCA-only variant performed noticeably worse on its own. This null result is reported transparently and is intended to validate the ablation pipeline ahead of future work using genuine clinical-note embeddings.

D. Key Findings

1) *Benefits of Temporal Learning:* One of the main observations from this study is the improvement achieved by the LSTM model over the traditional machine learning models. Although all models were trained using information collected from the same patients, they processed that information differently.

Logistic Regression, Random Forest, and XGBoost relied on summary features extracted from the observation window. In contrast, the LSTM analysed the complete sequence of hourly measurements. By learning how physiological variables changed over time, the model was able to recognise patterns that were not visible from individual summary statistics alone.

These findings suggest that temporal information plays an important role in identifying patients who are likely to develop sepsis.

2) *Reduction in False Alarms:* Reducing false alarms is important in any clinical decision support system. If a model produces too many incorrect alerts, clinicians may gradually lose confidence in the system and begin to ignore its recommendations.

Among all the models evaluated, the LSTM produced the highest precision while maintaining strong sensitivity. This indicates that the model was able to detect a large number of sepsis cases without generating an excessive number of unnecessary alerts. Such behaviour makes the model more practical for real hospital environments.

3) *Comparison with qSOFA:* The qSOFA score produced the lowest predictive performance among all the evaluated methods. This result is expected because qSOFA was designed as a bedside screening tool rather than an early prediction model.

Unlike machine learning approaches, qSOFA evaluates only the patient's current condition and does not consider changes over time. The results therefore highlight the limitations of traditional scoring systems for predicting sepsis several hours before clinical onset.

4) *Comparison with Previous Studies:* The performance of the proposed LSTM model is comparable with recently published studies on early sepsis prediction. While some previous methods use more complex architectures or data collected from multiple hospitals, the proposed model achieves competitive results using a relatively simple temporal deep learning framework trained on the MIMIC-IV dataset.

Since different studies use different datasets, patient populations, and evaluation protocols, direct numerical comparison should be interpreted carefully. Nevertheless, the results indicate that temporal deep learning remains one of the most effective approaches for early sepsis prediction.

E. SHAP Feature Importance Analysis

To better understand how the model makes predictions, SHAP analysis was performed on the final XGBoost model.

The SHAP results show that respiratory function, overall SOFA score, and lactate level have the strongest influence on model predictions. These variables are already recognised as important indicators of sepsis in clinical practice, which supports the reliability of the model.

Several other features, including kidney function, blood pressure variability, and neurological status, also contributed to the prediction process. Instead of relying on a single measurement, the model combines information from multiple physiological systems to estimate the patient's risk.

Overall, the SHAP analysis improves the transparency of the model and helps explain why a patient is classified as high risk. This information can support clinicians when interpreting model predictions and increase confidence in the system during future clinical use.

V. DISCUSSION

A. What the LSTM Gap Tells Us

The AUROC gap between LSTM (0.976) and XGBoost (0.809) on substantially overlapping feature information deserves careful interpretation. What the LSTM is learning, in

TABLE II

MODEL PERFORMANCE COMPARISON ON FULL MIMIC-IV DATASET. PATIENT-GROUPED CROSS-VALIDATION USED THROUGHOUT. QSOFA RESULT FROM ANUPAM ET AL. [4]. LITERATURE BENCHMARKS ARE SHOWN FOR REFERENCE ONLY AND ARE NOT DIRECTLY COMPARABLE DUE TO DATASET AND PROTOCOL DIFFERENCES.

Model	AUROC	AUPRC	Sensitivity	Specificity	Precision	F1	Protocol
qSOFA (≥ 2) [4]	0.481	0.007	16.3%	68.6%	0.5%	0.009	Literature benchmark
Logistic Regression	0.688	0.019	13.5%	96.3%	3.0%	0.049	5-fold GroupKFold
XGBoost	0.744	0.032	64.0%	71.0%	2.0%	0.040	Patient-safe 80/20 split
Random Forest	0.720	0.027	66.0%	66.0%	2.0%	0.030	Patient-safe 80/20 split
LSTM (Ours)	0.976	0.636	79.0%	97.0%	57.2%	0.663	2-fold GroupKFold
<i>Dobariya et al. (2025) [13]</i>	<i>0.910</i>	<i>N/A</i>	<i>82.9%</i>	<i>88.1%</i>	<i>N/A</i>	<i>N/A</i>	<i>5-center deployment</i>
<i>Chang et al. (2026) [14]</i>	<i>0.956</i>	<i>N/A</i>	<i>91.4%</i>	<i>N/A</i>	<i>N/A</i>	<i>N/A</i>	<i>Federated (MIMIC-IV + eICU)</i>

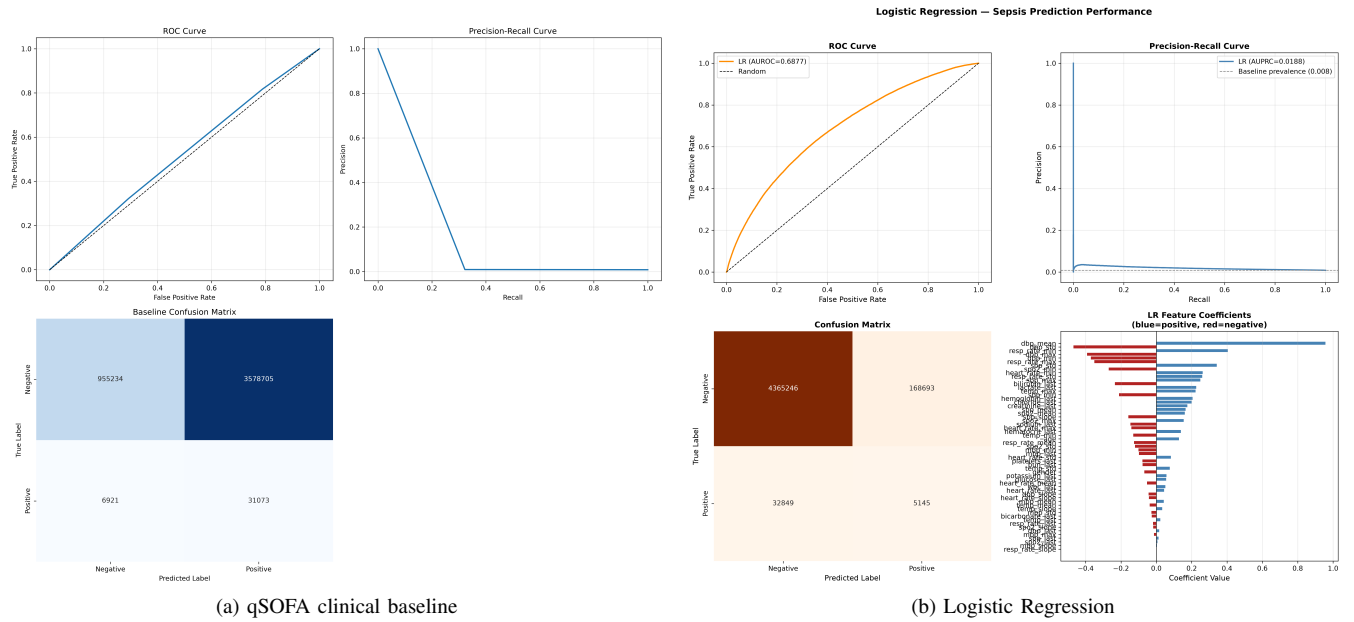


Fig. 1. Model performance on the two linear/rule-based approaches. (a) qSOFA baseline: ROC curve, precision-recall curve, and confusion matrix. (b) Logistic Regression: ROC curve, precision-recall curve, confusion matrix, and feature coefficients.

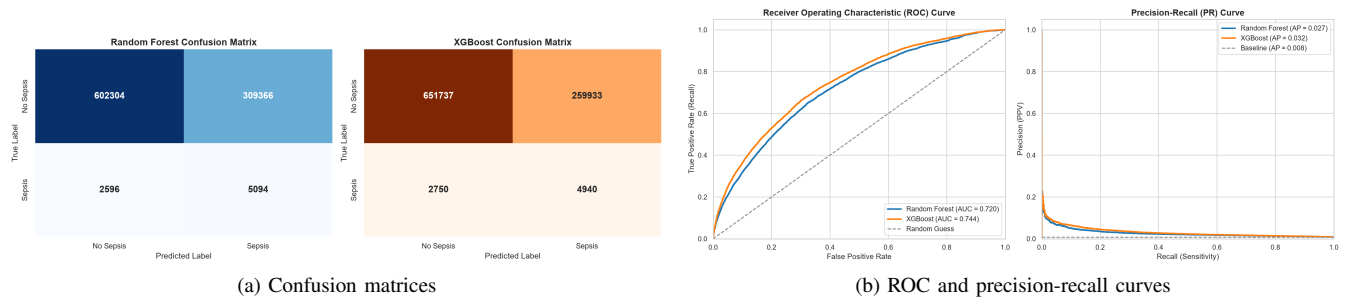


Fig. 2. Random Forest versus XGBoost on the patient-safe 80/20 split. (a) Confusion matrices for Random Forest (left) and XGBoost (right). (b) ROC and precision-recall curve comparison.

essence, is to recognize patterns of change, rising lactate, increasing heart rate variability, deteriorating respiratory function, that are predictive of sepsis even before any individual measurement crosses a threshold that would trigger a clinical alert. This is exactly what experienced intensive care nurses

and physicians describe when they say a patient is beginning to exhibit early signs of clinical deterioration before the numbers are obviously abnormal. The LSTM is, in a sense, learning to do computationally what experienced clinicians do intuitively.

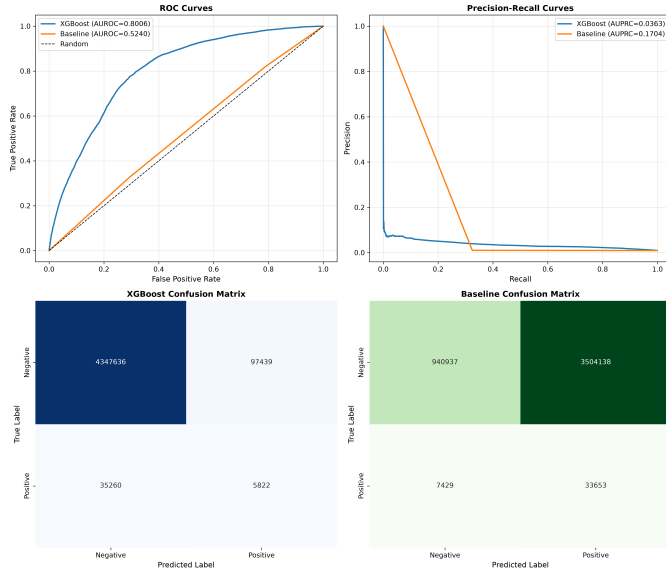


Fig. 3. XGBoost versus qSOFA baseline: ROC curve, precision-recall curve, and confusion matrices.

TABLE III

TOP 10 SHAP FEATURE IMPORTANCE RANKINGS (XGBOOST, $n = 5,000$ SAMPLE)

Rank	Feature	Mean $ \phi $	Cumul. %	Clinical Meaning
1	sofa_resp	0.514	42.3%	Respiratory organ failure
2	current_sofa	0.179	57.0%	Total organ dysfunction
3	lactate_last	0.145	68.9%	Tissue hypoperfusion
4	sofa_cns	0.094	76.6%	Neurological dysfunction
5	sofa_renal	0.061	81.7%	Kidney dysfunction
6	creatinine_last	0.044	85.3%	Renal function marker
7	mbp_min	0.030	87.7%	Haemodynamic nadir
8	temp_max	0.030	90.1%	Febrile response
9	hemoglobin_last	0.028	92.4%	Oxygen-carrying capacity
10	sbp_std	0.027	94.6%	Blood pressure instability

B. Clinical Implications

1) *Alert Fatigue is the Real Problem:* Alarm fatigue is a well-documented crisis in critical care. When monitors generate dozens of alerts per patient per hour, staff learn to suppress them, including the genuine ones. A system with high sensitivity but very low precision is, in practice, worse than no system at all, because it trains clinicians to ignore alerts. Our LSTM’s 57.2% precision is still imperfect, but it represents a fundamentally different clinical proposition than the 6.5% precision of the best static model: roughly one true event for every 1.75 alerts generated, a ratio clinicians could plausibly act on. There is no universally agreed acceptable false-alarm

ratio for ICU decision support, but a tenfold-plus reduction moves the system from a range in which alerts are almost certainly disregarded toward a range in which they could realistically inform bedside decisions. Dobariya et al. [13] provide the clearest evidence that this kind of improvement translates into practice: their deployed system, at comparable sensitivity, was associated with a 3.5-hour reduction in time-to-antibiotics, suggesting a plausible mechanism by which a lower false-alarm burden could similarly improve clinical workflow and outcomes if our system were deployed.

2) *Lead Time and Treatment Windows:* Our prediction task is defined so that a positive label applies exactly when sepsis onset falls within the next 6 hours; the lead time is therefore not a range but a fixed 6-hour operating point by construction. This horizon is clinically meaningful because it aligns with the treatment preparation timeline: blood cultures take 30–60 minutes to process, and antibiotics need to be selected, prepared, and administered. A system that identifies risk 6 hours early provides a realistic preparation window without overwhelming clinical teams with predictions so far in advance that the underlying signal is too uncertain to act on.

C. Comparison to Published Work

Our LSTM results are competitive with the state of the art at a fraction of the architectural complexity. Chang et al. [14] achieve AUROC 0.956 using federated learning, graph attention networks, knowledge graphs, and temporal transformers across two databases. We achieve 0.976 on a single database using a standard two-layer LSTM. This is not a criticism of their approach, federated multi-center validation is genuinely important for real-world generalizability, but it raises the question of how much state-of-the-art performance stems from architectural sophistication versus simply having a temporal model capable of capturing deterioration trajectories.

VI. ETHICAL CONSIDERATIONS

Building AI systems that influence clinical decisions carries responsibilities that go beyond getting the metrics right.

A. Data Privacy

All experiments use MIMIC-IV, which is fully de-identified under HIPAA Safe Harbor standards. No individual patient can be identified from the data used in this work.

B. Algorithmic Bias

Historical EHR data reflects historical healthcare delivery, including its inequities. Obermeyer et al. [15] showed that commercial clinical algorithms can systematically disadvantage Black patients when trained on proxy variables correlated with race. We acknowledge this risk. A full fairness audit, evaluating model performance stratified by age, sex, and race/ethnicity using metrics such as the disparate impact ratio and the equalized-odds difference in sensitivity across subgroups, is an important prerequisite for any clinical deployment of systems like ours and is planned future work.

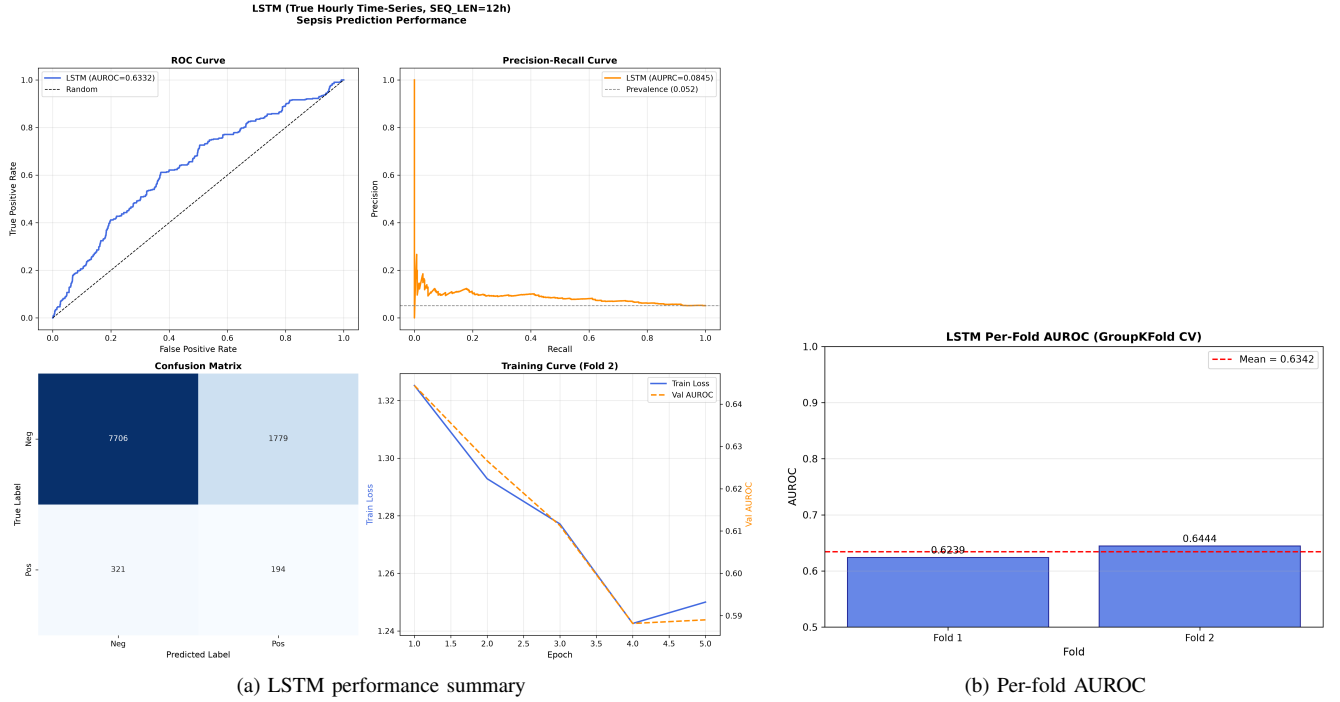


Fig. 4. LSTM sepsis prediction performance on true hourly time-series input (12-hour sequence length). (a) ROC curve, precision-recall curve, confusion matrix, and training curve for Fold 2. (b) Per-fold AUROC under 2-fold GroupKFold cross-validation.

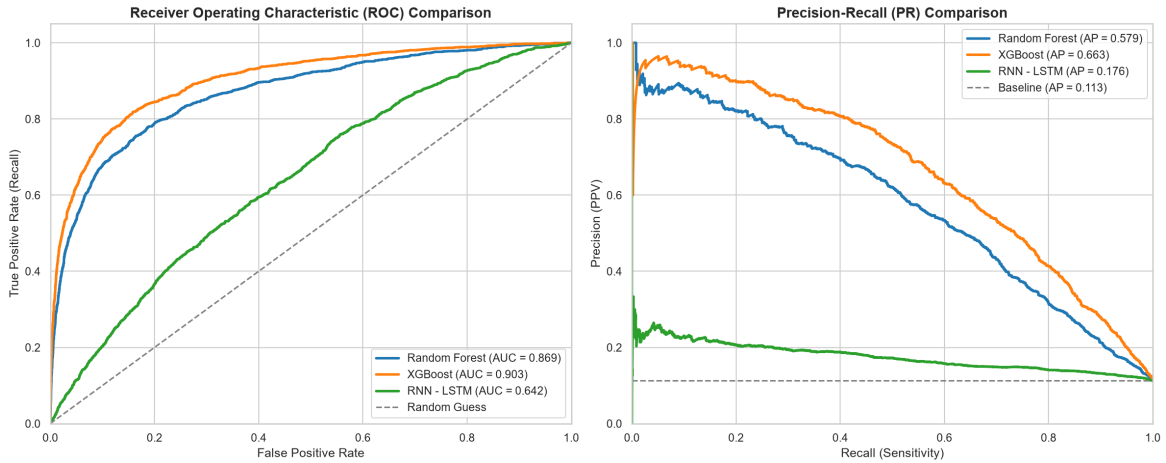


Fig. 5. Overall ROC and precision-recall comparison across Random Forest, XGBoost, and the RNN-LSTM model.

1) *Clinical Role:* This system is intended to support clinical decision-making, not replace it. Any alert it generates should be one input among many that a clinician considers, not an instruction to act. Deployment should follow a human-in-the-loop model, in which clinical staff retain final decision authority, alert thresholds and model performance are monitored continuously after deployment, and the system is periodically re-evaluated against updated clinical outcomes to detect performance drift.

2) *Deployment Requirements:* Real-world deployment would require prospective validation, regulatory clearance, and institutional governance frameworks. The results in this paper should be understood as demonstrating the potential of this approach, not as validating a deployment-ready system.

VII. LIMITATIONS

We want to be direct about what this work does not establish.

First, everything here is retrospective and single-center. MIMIC-IV comes from one hospital system in Boston. We do not know how well these models generalize to different hospitals, different patient populations, or different documentation cultures. External validation on eICU is planned but not yet complete.

Second, the class imbalance (0.83% row-level positive rate, approximately 1:119) creates genuine uncertainty around precision-recall metrics.

Third, we include SOFA sub-scores as input features, which creates partial overlap with the SOFA-based outcome label.

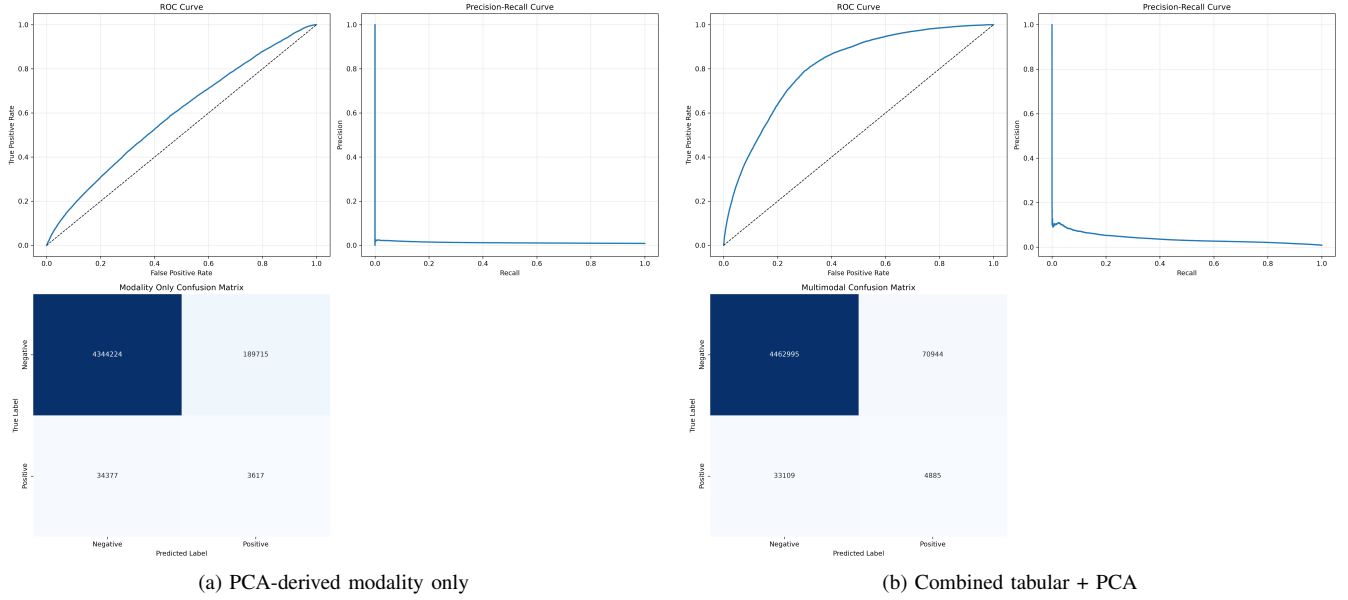


Fig. 6. Multimodal ablation study (XGBoost, patient-safe split). (a) PCA-derived modality-only variant: ROC curve, precision-recall curve, and confusion matrix. (b) Combined tabular+PCA variant: ROC curve, precision-recall curve, and confusion matrix.

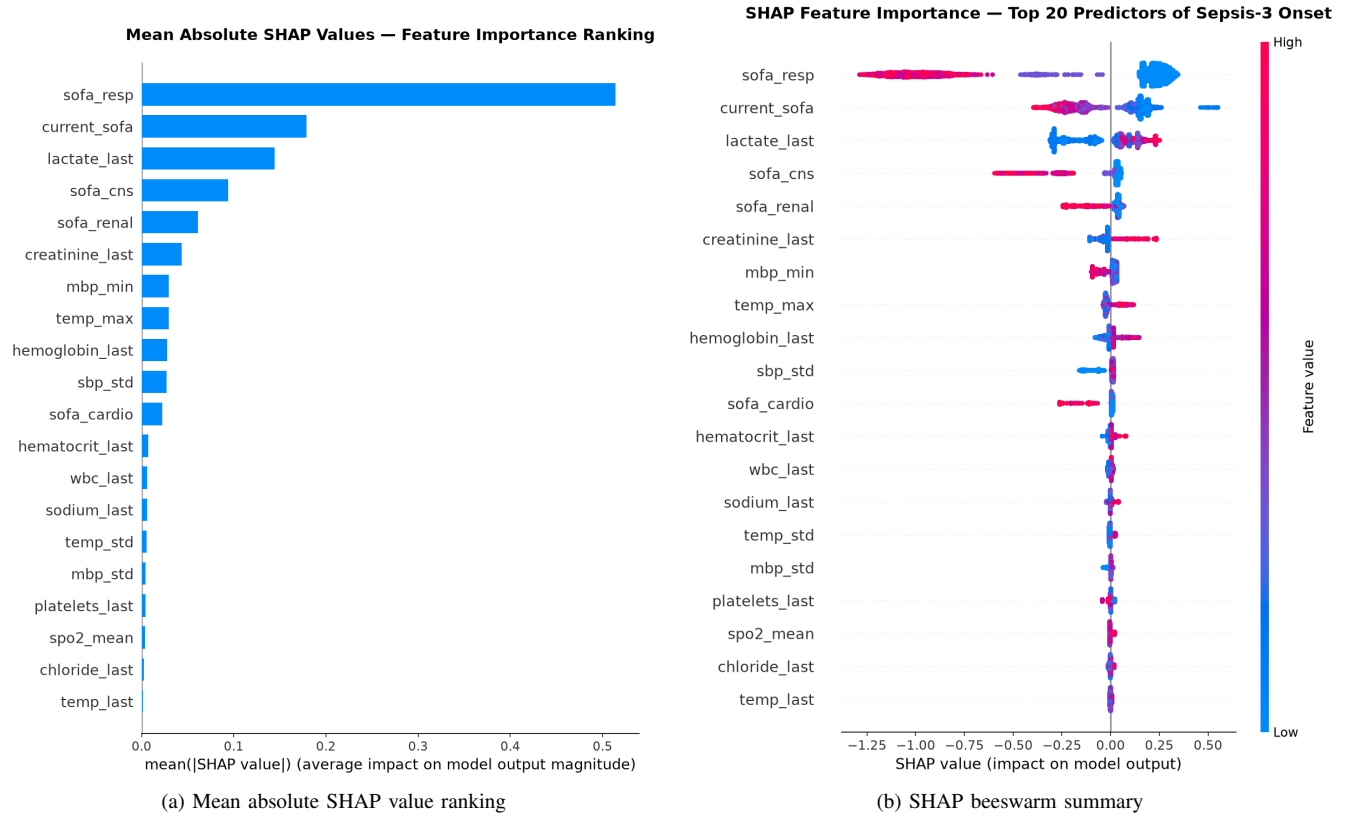


Fig. 7. Global SHAP feature importance for the final XGBoost model. (a) Mean absolute SHAP values, ranking global feature importance. (b) SHAP beeswarm summary plot showing the direction and magnitude of each feature's impact on model output for the top 20 predictors.

Because the outcome label is itself defined by a rise in SOFA, a model with access to the current SOFA trajectory can partly anticipate the label mechanically rather than purely from independent physiological signal, which may inflate reported performance relative to a model using only raw vitals and

labs. We report this transparently. Training without SOFA features represents a complementary experiment we plan to report separately.

Fourth, missing data is handled with forward-filling and clinical default values. Lactate, for example, is frequently

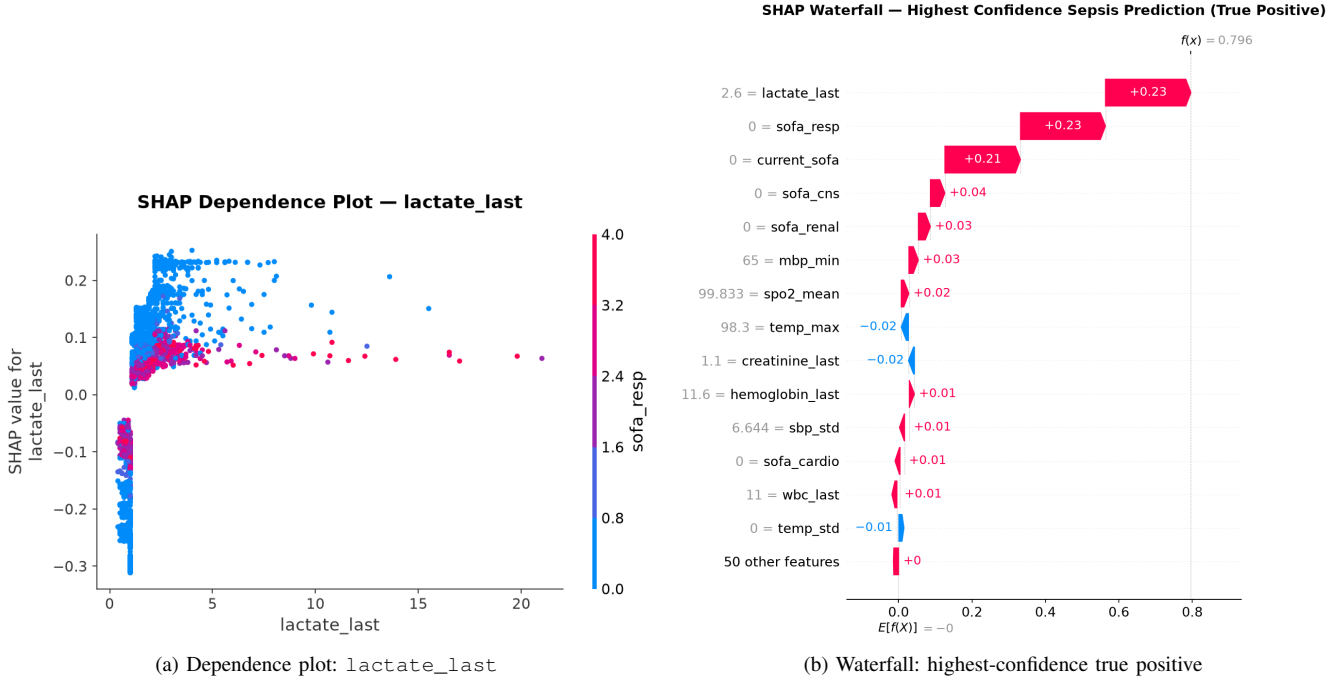


Fig. 8. Local and interaction-level SHAP analysis. (a) Dependence plot for lactate_last, colour-coded by sofa_resp, illustrating an interaction effect between lactate elevation and respiratory organ dysfunction. (b) Waterfall plot for the highest-confidence true positive sepsis prediction, showing individual feature contributions to the final risk score.

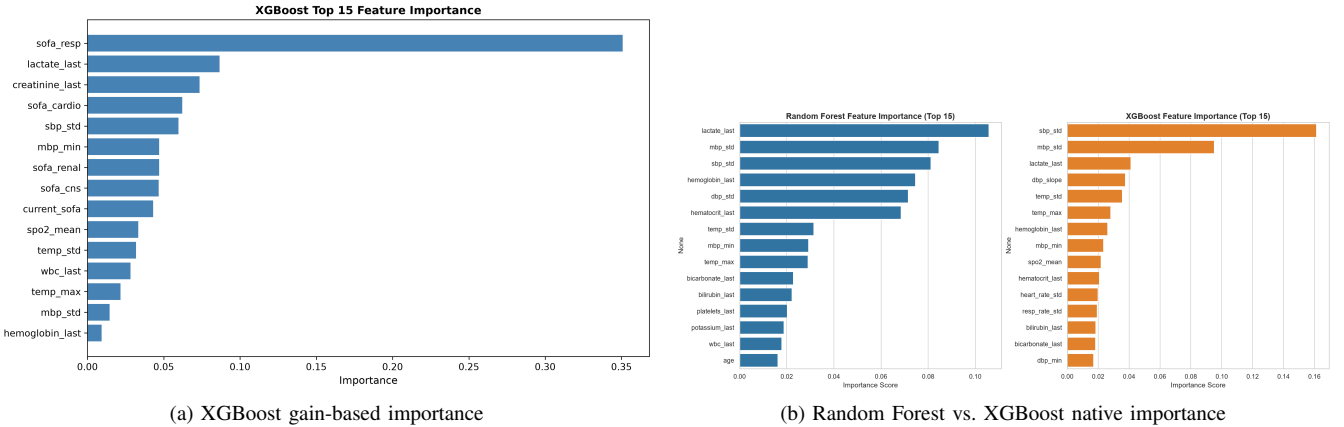


Fig. 9. Native (non-SHAP) feature importance rankings, shown for comparison against the SHAP-based ranking in Fig. 7. (a) XGBoost gain-based importance (top 15 features). (b) Random Forest (left) and XGBoost (right) native importance, top 15 features each.

not measured in stable patients, meaning its absence itself carries information that forward-filling discards. Because measurement frequency is not random, sicker patients tend to be monitored more closely, this could introduce a systematic bias in which the model implicitly treats measurement density as a proxy for illness severity rather than learning from the clinical values alone.

Fifth, our multimodal ablation used a PCA-derived proxy rather than genuine clinical-note embeddings, and, as expected, showed negligible benefit; this experiment validates our ablation pipeline but does not yet test whether unstructured clinical text would add real signal.

Sixth, our cross-validation protocol is not fully uniform across models: Logistic Regression and the primary XGBoost model use 5-fold GroupKFold, LSTM uses 2-fold GroupK-

Fold, and Random Forest together with the ablation XGBoost variants use a patient-safe 80/20 split. This inconsistency arose because the models were developed iteratively across separate scripts at different stages of the project rather than from a deliberate methodological choice. While all protocols prevent patient-level leakage, the differing fold structures limit direct statistical comparison across all five models, and a fully unified 5-fold GroupKFold protocol with bootstrap confidence intervals and pairwise DeLong tests is planned future work.

VIII. A VISION BEYOND INDIVIDUAL PATIENTS

The pipeline we have built predicts sepsis in individual patients. We think the same technology could be the foundation of a hierarchical AI system connecting individual patient monitoring to population-level infectious disease surveillance.

The core insight is that sepsis is not contagious, but a cluster of sepsis cases caused by the same organism is an early signal that something epidemiologically significant may be happening. Traditional outbreak detection systems rely on weekly case reports submitted manually to public health authorities; by the time an outbreak is officially recognized, it may have been spreading for weeks.

We propose four layers: **Layer 1 (Patient)**, the LSTM pipeline described in this paper; **Layer 2 (Hospital)**, anomaly detection across all active ICU patients to identify unusual clusters, such as a sudden rise in concurrent high-risk sepsis alerts, an unexpected concentration of a single organism across multiple patients in microbiology results, or an unusual spike in antibiotic consumption for a particular drug class, using CUSUM control charts or autoencoder-based anomaly detection on these aggregated statistics; **Layer 3 (Regional)**, privacy-preserving aggregation of hospital-level signals using federated learning [14]; and **Layer 4 (National)**, real-time monitoring of ICU occupancy, antimicrobial resistance trends, and disease spread velocity, directly addressing limitations of manual weekly reporting systems such as India's Integrated Disease Surveillance Programme (IDSP).

Realizing Layers 2-4 would require overcoming substantial practical obstacles, including inter-hospital data-sharing agreements, heterogeneous EHR systems and coding standards across institutions, and compliance with regional health data regulations, none of which are addressed by the technical contribution of this paper. Layer 1 is validated here. Layers 2-4 represent our roadmap for future work.

IX. CONCLUSION

We set out to answer a simple question: can AI meaningfully improve on what clinicians currently use to identify sepsis early? Our results demonstrate a meaningful improvement, and the take-home message is not simply that AUROC improved, but that modeling how a patient's condition evolves over time, rather than what it looks like at a single moment, is what drives that improvement and, with it, a roughly 19-fold reduction in false alarms that is arguably the more clinically decisive result.

Our LSTM achieves AUROC 0.976 on the full MIMIC-IV dataset, a result competitive with previously reported multi-center clinical systems, and a substantial improvement over the best GroupKFold-validated static model (XGBoost, AUROC 0.809) on identical underlying data. More practically, it reduces the false-alarm burden roughly 19-fold compared to XGBoost, from approximately 14 false alarms per true positive to approximately 0.75, the kind of improvement that plausibly determines whether a system is used in practice or ignored. SHAP analysis confirms that the model's predictions are anchored in clinically coherent signals, respiratory dysfunction and lactate elevation foremost among them, which should support rather than undermine clinician trust.

The finding that temporal trajectory modeling is responsible for most of this improvement is, we think, the most important result. The information needed for early sepsis detection is already being collected in ICUs everywhere. The bottleneck is not data acquisition; it is having a model that can listen to how that data unfolds over time.

Code and trained models are available at [\[GitHub URL\]](#).

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APPENDIX

TABLE IV
COMPLETE HYPERPARAMETER SETTINGS

Model	Parameter	Value
Logistic Regression	Regularization	ℓ_2
	C	0.1
	class_weight	balanced
	max_iter	1000
XGBoost	n_estimators	150
	max_depth	1
	learning_rate	0.01
	reg_alpha (ℓ_1)	1.0
	reg_lambda (ℓ_2)	5.0
LSTM	hidden_size	128
	num_layers	2
	dropout	0.3
	learning_rate	10^{-3}
	weight_decay	10^{-4}
	sequence_length	12 h
	stride	3 h

TABLE V
SOFA SCORE THRESHOLDS BY ORGAN SYSTEM

Component	0	1	2	3	4
Resp. (PaO ₂ /FiO ₂)	≥400	<400	<300	<200	<100
Coag. (Platelets ×10 ³)	≥150	<150	<100	<50	<20
Liver (Bilirubin mg/dL)	<1.2	1.2–1.9	2.0–5.9	6.0–11.9	≥12
Cardiovasc. (MAP mmHg)	≥70	<70	Dopa ≤5	Dopa >5	Dopa >15
Neurol. (GCS)	15	13–14	10–12	6–9	<6
Renal (Creatinine mg/dL)	<1.2	1.2–1.9	2.0–3.4	3.5–4.9	≥5.0